# COURSE OUTLINE

1. **Course Title:** Digital Image Processing
2. **Course Code:** CSE 4245
3. **Credit Hours:** 3.00
4. **Semester:** Winter 2026
5. **Level / Term:** Level 4, Term II
6. **Faculty:** Faculty of Electrical and Computer Engineering
7. **Department:** Department of Computer Science and Engineering
8. **Program:** B.Sc. Engineering in Computer Science and Engineering
9. **Synopsis of the Curriculum**

   Digital Image Processing (DIP) is a specialty-related elective/optional course for the Engineering students. This course introduces digital image processing. It focuses on the theory and algorithms underlying a range of tasks including image acquisition and formation, enhancement, restoration, compression, segmentation and object recognition, etc.

10. **Type of Course (Core/Elective):** Elective
11. **Prerequisite(s) (if any):** None
12. **Name of the instructor(s) with contact details and office hours:**

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|---|---|
| Name: Dr. S.M. Jahangir Alam<br>Designation: Professor<br>Contact no: 01766128002<br>Email: jahangir@baust.edu.bd<br>Counselling Hour: 10:50-11:00 | Name: N/A<br>Designation: N/A<br>Contact no: N/A<br>Email: N/A<br>Counselling Hour: N/A |

## 13. CO – PO mapping and Bloom's Taxonomy level:

| CO No. | CO Statement | POs | Bloom's Taxonomy | | |
|---|---|---|---|---|---|
| | | | C | A | P |
| CO1 | ***Explain*** *the fundamentals of digital image processing, image formation, representation, Quantization, and pixel Connectivity.* | *PO1* | *C2* | - | - |
| CO2 | ***Apply*** *spatial and frequency domain techniques for image enhancement and restoration, Quantization, Pixel Connectivity.* | *PO1* | *C3* | - | - |
| CO3 | ***Analyze and Implement*** *image segmentation, edge detection, and morphological operations.* | *PO5* | *C3* | - | - |
| CO4 | ***Evaluate and Design*** *images using feature extraction and classification methods, Applications.* | *PO4* | *C5* | - | *P1* |

## 14. Justification of CO-PO Mapping

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| CO1-PO1 | Introduces fundamental knowledge of image formation and analysis, fostering analytical thinking. |
| CO2-PO1 | Builds understanding and ability to apply fundamental image processing techniques for improving and restoring images. |
| CO3-PO5 | Develops the ability to apply practical image processing techniques using appropriate tools and methods. |
| CO4-PO11 | Involves investigating and interpreting results to support problem-solving and decision-making. |

## 15. CO - Knowledge Profile - Complex Engineering Problem Solving - Complex Engineering Activities Mapping

| Course Outcome (CO) | Knowledge Profile (K) | Complex Problem Solving (P) | Complex Engineering Activities (A) |
|---|---|---|---|
| *CO1* | *K1- K4* | - | - |
| *CO2* | *K3-K6* | - | - |
| *CO3* | *K6* | - | - |
| *CO4* | *K3-K6, K8* | *P1* | - |

## 16. Percentages of Assessment Methods

| Method | Percentage | CO |
|---|---|---|
| Class Performance | 5% | |
| Class Test | 15% | CO1, CO2 |
| Assignments | 5% | CO3, CO4 |
| Mid-term Examination | 15% | CO1, CO2 |
| Final Examination | 60% | CO2, CO3 |
| Total | 100% | |

## 17. Weekly distribution of course contents and assessment methods

| Week | Topics | COs | Assessment Method(s) |
|---|---|---|---|
| 1. | **Digital Image Fundamentals:**<br>• Human Visual System: Pixels & Resolution, Knowledge of the human eye's structure and how it perceives images, Image Acquisition: How images are captured by sensors and converted into digital form, Image Contrast. | CO1 | Class Test 1, Mid-term Exam |
| 2. | **Digital Image Fundamentals:**<br>• Key Stages in DIP, Digital Images. Image Representation, Image Classification, Image Description, Noise in Image, Parametric Method in Image Processing, Noise in Images, Entropy, Image Quality.<br>• Regions & Boundaries of an Image, Digital Depth (or Curve) Image, Some basic relationships between pixels, paths, Pixel Connectivity and graph.<br>• Linear and Nonlinear operations, Image Formation Model, Image sampling and quantization.<br>• K-means-based or Grayscale Quantization, Max-Lloyd Algorithm, Image Interpolation, Applications. | CO1, CO2 | Class Test 1, Mid-term Exam, Final Exam |
| 3. | **Image Enhancement**<br>• Histogram and Contrast stretching, Histogram equalization. Image matching.<br>• Spatial Filtering, Correlation & Convolution, Kernel, LTI, FIR.<br>• Smoothing Filters (e.g., Mean Filter, Median Filter, Gaussian Filter): Reduce noise by averaging neighboring pixels.<br>• Spatial Filtering and Kernels for Smoothing Intensities, Comparisons between Filtering in Spatial and Frequency Domain. | CO2 | Class Test 1, Mid-term Exam, Final Exam |
| 4. | **Image Enhancement**<br>• Spatial Filtering and Kernels for Smoothing Intensities, Box Filter Kernels, The Laplacian Filter, Unsharp Masking, Wavelet Transform and sub-band Decomposition, Image Enhancement | CO2 | Mid-term Exam, Final Exam |

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| | in Frequency Domain.<br>• Frequency Domain Techniques, 2D Fourier transform, filtering in the frequency domain, and their applications, Wavelet Transform and sub-band Decomposition. | | |
| 5. | **Color Image Processing**<br>• **Color Models:** CIE, RGB, CMYK, HSI, HSV, and other color spaces.<br>• **Color Transformations:** Color balancing, color correction, and pseudocolor processing.<br>• **Color Correction and Balancing**: Adjust color tones and illumination. | CO1 | Class Test 2, Mid-term Exam, Final Exam |
| 6. | **Image Transforms:**<br>• Geometric transformations (2D and 3D scaling, translation, rotation, affine transform), Intensity Transformations & Spatial Filtering, Intensity Transformations Functions.<br>• DFT (Discrete Fourier Transform): Properties, applications, and relationship to spatial domain techniques.<br>• Other Transforms: Walsh-Hadamard Transform (WHT), 1-D and 2-D Karhonen Loeve (KLT) or principal component analysis (PCA). | CO3 | Class Test 2, Final Exam |
| | **Midterm Exam** | | |
| 7. | **Image Segmentation**<br>• **Thresholding:** Global, local, adaptive thresholding techniques, Gaussian Adaptive Thresholding and Outsu's Algorithm.<br>• Outsu's Algorithm, Optimum Global Thresholding Using Otsu's Method.<br>• Point, Line operations, Edge detection and Linking (e.g., Sobel, Prewitt, Roberts, Canny): Detects object boundaries based on intensity gradients, Local Patterns of Gradients (LPG).<br>• Laplacian Edge Detection, LoG/ Marr-Hildreth edge detector, DoG. | CO3 | Class Test 2, Final Exam, Assignment |
| 8. | **Image Segmentation**<br>• Image Segmentation, Region-Based Segmentation, Region Growing and Segmentation: Region growing, Border Detection as Graph Searching, Region Merging, and other techniques, Region Splitting and Merging algorithms, and other techniques.<br>• **Watershed Algorithm**: Treats image as topographic surface to segment regions. | CO2 | Class Test 2, Final Exam |
| 9. | **Image Restoration**<br>• **Noise Models:** Understanding different types of noise and their characteristics.<br>• **Median Filtering**: Removes salt-and-pepper noise by replacing pixel with median of neighbors.<br>• **Restoration Techniques:** Inverse filtering, Wiener filtering, and other methods for removing noise and distortions. | CO1 | Class Test 3, Final Exam |

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| | • **Inverse Filtering**: Attempts to reverse degradation by inverting the degradation function.<br>• **Wiener Filtering**: Minimizes mean square error between restored and original image. | | |
| 10. | **Image Compression**<br>• **Fundamentals:** Lossless and lossy compression techniques, compression ratio, and redundancy.<br>• **Compression Models:** Predictive models, transform coding, JPEG, MPEG, H.261, H.263, Video compression and other techniques.<br>• **Run-Length Encoding (RLE)**: Compresses sequences of repeated pixels. | CO2 | Class Test 3, Final Exam, Assignment |
| 11. | **Image Compression**<br>• Image Compression Fundamentals: Lossless and lossy compression techniques, compression ratio, and redundancy, Run-Length Encoding (RLE), VLC, Huffman Coding.<br>• Arithmetic Coding, Golomb Coding, LZW.<br>• **Transform Coding** (e.g., DCT in JPEG, Wavelet Transform): Converts image to frequency domain, then quantizes coefficients.<br>• Compression Models: Predictive models, transform coding, JPEG, MPEG, H.261, H.263, Video compression and other techniques.<br>• **Predictive Coding**: Encodes differences between pixels, Entropy-based schemes, Transform-based encoding, Predictive encoding and DPCM, Huffman coding.<br>• **Vector Quantization**: Groups block of pixels and replaces with closest codeword. | CO2 | Final Exam |
| 12. | **Morphological Processing**<br>• Logic Operations, Morphological Operations: Erosion, dilation.<br>• Opening, closing, and their combinations, Hit-or-Miss Transform, Morphological Gradient, Top-Hat Transform. **Opening and Closing**: Remove small objects or fill holes.<br>• Skeletonization: Extracts medial axis of shapes, Skeletonization Algorithm,<br>• **Applications:** Feature extraction, noise reduction, and other tasks. | CO2 | Final Exam |
| 13. | **Object Feature Extraction and Pattern Recognition**<br>• **Feature Extraction:** Techniques for extracting relevant features from images.<br>• **Edge and Corner Detectors** (e.g., Harris Corner Detector).<br>• **SIFT, SURF**: Extract distinctive keypoints for matching.<br>• **Template Matching**: Detect known patterns.<br>• **Pattern Recognition:** Classification methods and their applications, Standard linear and Bayesian classifiers, supervised vs unsupervised classification, classification | CO3, CO4 | Final Exam, Assignment |

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| | performance index.<br>• **Machine Learning / Deep Learning**: CNNs for image classification and recognition.<br>• Contour and shape dependent feature extraction, textural features, region-based and feature-based segmentation. | | |
| 14. | **Revision & Wrap-up** | | |
| | **PL: June 7-20** | | |
| | **Final Exam: 21 June-25 July** | | |
| | **RIB: 26 July-Aug 1** | | |

## 18. Text Book

1. Digital Image Processing. Fourth Edition. Rafael C. Gonzalez & Richard E.

## 19. References

1. Digital Image Processing using MATLAB. Second Edition. Rafael C. Gonzalez & Richard

| **Prepared by:** | **Checked by:** | **Approved by:** |
|---|---|---|
| Signature: _____<br><br>Name: Prof. Dr. S.M. Jahangir Alam<br>Department: CSE<br>Date:<br>Counseling hour: 10.50-12.30 | Signature: _____<br><br>Name: Md. Al-Hasan<br>OBE Program Coordinator,<br>Department of CSE<br>Date: | Signature: _____<br><br>Name: Md Nakib Hayat Chowdhury, PhD<br>HoD,<br>Department of CSE<br>Date: |

| Lec | Date | Topics | COs | Assessment Method(s) |
|---|---|---|---|---|
| 1 | 3/2 | **Digital Image Fundamentals:** Human Visual System: Pixels & Resolution, Knowledge of the human eye's structure and how it perceives images, Image Acquisition: How images are captured by sensors and converted into digital form, Image Contrast. | CO1 | CT1, MT Exam |
| 2 | 4/2 | Key Stages in DIP, Digital Images. | | |
| 3 | 5/2 | Image Representation, Image Classification, Image Description, Noise in Image, Parametric Method in Image Processing, Noise in Images, Entropy, Image Quality. | | |
| 4 | 10/2 | Regions & Boundaries of an Image, Digital Depth (or Curve) Image, Some basic relationships between pixels, paths, Pixel Connectivity and graph. | CO1, CO2 | CT1, MT Exam, Final Exam |
| 5 | 18/2 | Linear and Nonlinear operations, Image Formation Model, Image sampling and quantization. | | |
| 6 | 19/2 | K-means-based or Grayscale Quantization, Max-Lloyd Algorithm, Image Interpolation, Applications. | | |
| 7 | 24/2 | **Image Enhancement** Histogram and Contrast stretching, Histogram equalization. | CO2 | CT2, MT Exam, Final Exam |
| 8 | 25/2 | Image matching. | | |
| 9 | 26/2 | Spatial Filtering, Correlation & Convolution, Kernel, LTI, FIR. | | |
| 10 | 3/3 | Smoothing Filters (e.g., Mean Filter, Median Filter, Gaussian Filter): Reduce noise by averaging neighboring pixels. | | |
| 11 | 4/3 | Spatial Filtering and Kernels for Smoothing Intensities, Comparisons between Filtering in Spatial and Frequency Domain. | | |
| 12 | 5/3 | Spatial Filtering and Kernels for Smoothing Intensities, Box Filter Kernels, The Laplacian Filter, Unsharp Masking, Wavelet Transform and sub-band Decomposition, Image Enhancement in Frequency Domain. | CO2 | CT2, MT Exam, Final Exam |
| 13 | 10/3 | Frequency Domain Techniques, 2D Fourier transform, filtering in the frequency domain, and their applications, Wavelet Transform and sub-band Decomposition. | | |
| 14 | 11/3 | **Color Image Processing** Color Models: CIE, RGB, CMYK, HSI, HSV, and other color spaces. | CO1 | CT2, MT Exam, Final Exam |
| 15 | 12/3 | Color Transformations: Color balancing, color correction, and pseudocolor processing. Color Correction and Balancing: Adjust color tones and illumination. | | |
| 16 | 31/3 | **Image Transforms:** Geometric transformations (2D and 3D scaling, translation, rotation, affine transform), Intensity Transformations & Spatial Filtering, Intensity Transformations Functions. | CO3 | CT2, Final Exam |

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| 17 | 1/4 | DFT (Discrete Fourier Transform): Properties, applications, and relationship to spatial domain techniques. | | |
| 18 | 2/4 | Other Transforms: Walsh-Hadamard Transform (WHT), 1-D and 2-D Karhonen Loeve (KLT) or principal component analysis (PCA). | | |
| | | **Midterm Exam: April 5-11** | | |
| 19 | 15/4 | **Image Segmentation**<br>Thresholding, Global, Adaptive/local Thresholding. | CO3 | CT2, Final Exam,<br><br>Assignment |
| | 16/4 | Gaussian Adaptive Thresholding. | | |
| 20 | 21/4 | Outsu's Algorithm, Optimum Global Thresholding Using Otsu's Method. | | |
| 21 | 22/4 | Point, Line detection. | | |
| 22 | 23/4 | Edge detection and Linking (e.g., Sobel): Detects object boundaries based on intensity gradients, Local Patterns of Gradients (LPG). | | |
| 23 | 28/4 | Edge detection (e.g., Prewitt, Roberts): Detects object boundaries based on intensity gradients. | | |
| 24 | 29/4 | Edge detection (e.g., Canny): Detects object boundaries based on intensity gradients. | | |
| 25 | 30/4 | Laplacian Edge Detection, LoG/ Marr-Hildreth edge detector, DoG. | | |
| 26 | 5/5 | Region-Based Segmentation, Region Growing and Segmentation: Region growing, Border Detection as Graph Searching, Region Merging, and other techniques, Region Splitting and Merging algorithms. | CO2 | CT3, Final Exam |
| 27 | 6/5 | Watershed Algorithm: Treats image as topographic surface to segment region. | | |
| 28 | 7/5 | **Image Restoration/Degradation**<br>Noise Models, Median Filtering for Removes salt-and-pepper noise by replacing pixel with median of neighbors. | CO1 | CT3, Final Exam |
| 29 | 12/5 | Inverse filtering, Wiener filtering, Minimum MSE using Wiener Filtering, and other methods for removing noise and distortions. | | |
| 30 | 13/5 | Image Processing Techniques by Modality, X-ray, CT, and MRI — Image Processing and applications. | | |
| 31 | 14/5 | **Image Compression**<br>Image Compression Fundamentals: Lossless and lossy compression techniques, compression ratio, and redundancy, Run-Length Encoding (RLE), VLC, Huffman Coding. | CO2 | Final Exam, Assignment |
| 32 | 19/5 | Arithmetic Coding, Golomb Coding, LZW. | | |
| | 20/5 | Image Transform Coding (e.g., DCT in JPEG, Wavelet Transform): Converts image to frequency domain, then quantizes coefficients. | | |
| 33 | 21/5 | Compression Models: Predictive models, transform coding, JPEG, MPEG, H.261, H.263, Video compression and other techniques. | | |

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| 34 | 26/5 | Predictive Coding: Encodes differences between pixels, Entropy-based schemes, Transform-based encoding, Predictive encoding and DPCM, Vector quantization, Huffman coding. | | |
| 35 | 27/5 | **Morphological Processing**<br>Logic Operations, Morphological Operations: Erosion, dilation. | CO2 | Final Exam |
| 36 | 28/5 | Opening, closing, and their combinations, Hit-or-Miss Transform, Morphological Gradient, Top-Hat Transform. | | |
| 37 | 2/6 | Skeletonization: Extracts medial axis of shapes, Skeletonization Algorithm, Applications: Feature extraction, noise reduction, and other tasks. | | |
| 38 | 3/6 | **Object Feature Extraction and Pattern Recognition**<br>Feature Extraction: Techniques for extracting relevant features from images.<br>Edge and Corner Detectors (e.g., Harris Corner Detector).<br>SIFT, SURF: Extract distinctive keypoints for matching.<br>Template Matching: Detect known patterns, Pattern Recognition: Classification methods and their applications. | CO3, CO4 | Final Exam, Assignment |
| 39-42 | 4-6/6 | Standard linear and Bayesian classifiers, supervised vs unsupervised classification, classification performance index, Machine Learning / Deep Learning: CNNs for image classification and recognition, Contour and shape dependent feature extraction, textural features, region-based and feature-based segmentation.<br>**Revision & Wrap-up** | | |
| | | **PL: June 7-20:** | | |
| | | **Final Exam: 21 June-25 July** | | |
| | | **RIB: 26 July-Aug 1** | | |